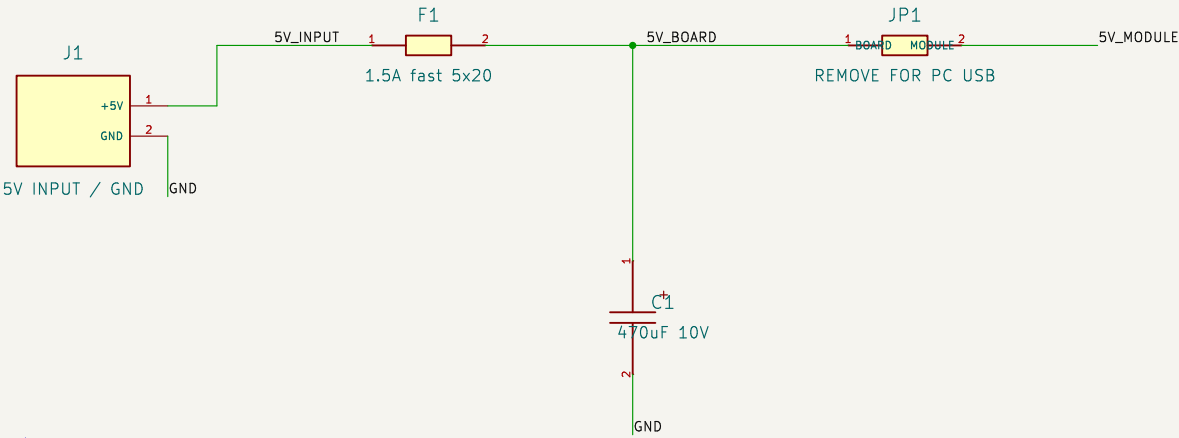


LINK COMMANDER / REV D / USB KEYBOARD + MOUSE TO PS/2

USB receiver -> RP2040 firmware -> PS/2 lines -> custom DIN cable -> Commander X16

1 / POWER

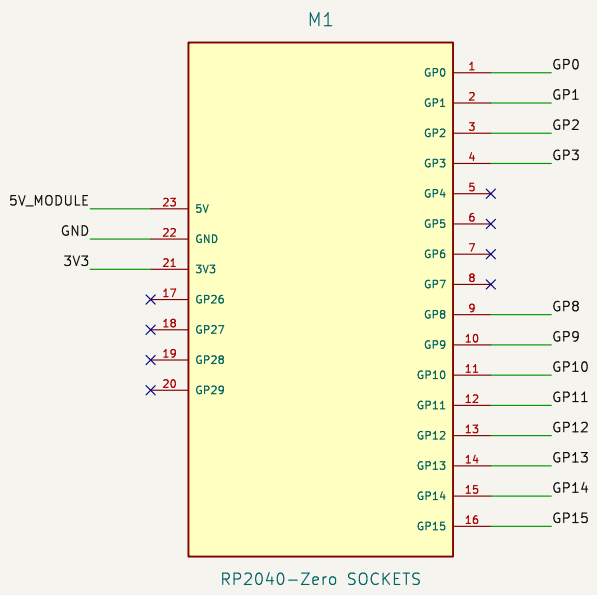
Regulated 5 V only. Remove JP1 before connecting PC USB.



J1: 5 V / 2 A supply
F1: 1.5 A fast fuse
USB ports turn on with external power.
X16 +5 V is used only for PS/2 pullups.

2 / REMOVABLE RP2040-ZERO

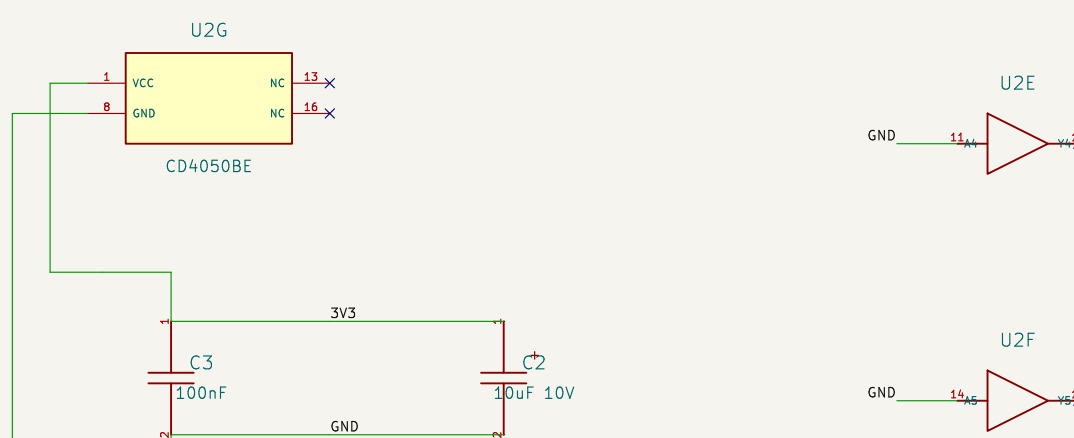
Plug-in module. No wires to underside pads.



GP0/1: USB keyboard D+/D-
GP2/3: USB mouse D+/D-
GPB-15: PS/2 drive and sense

3 / 3.3 V LOGIC POWER

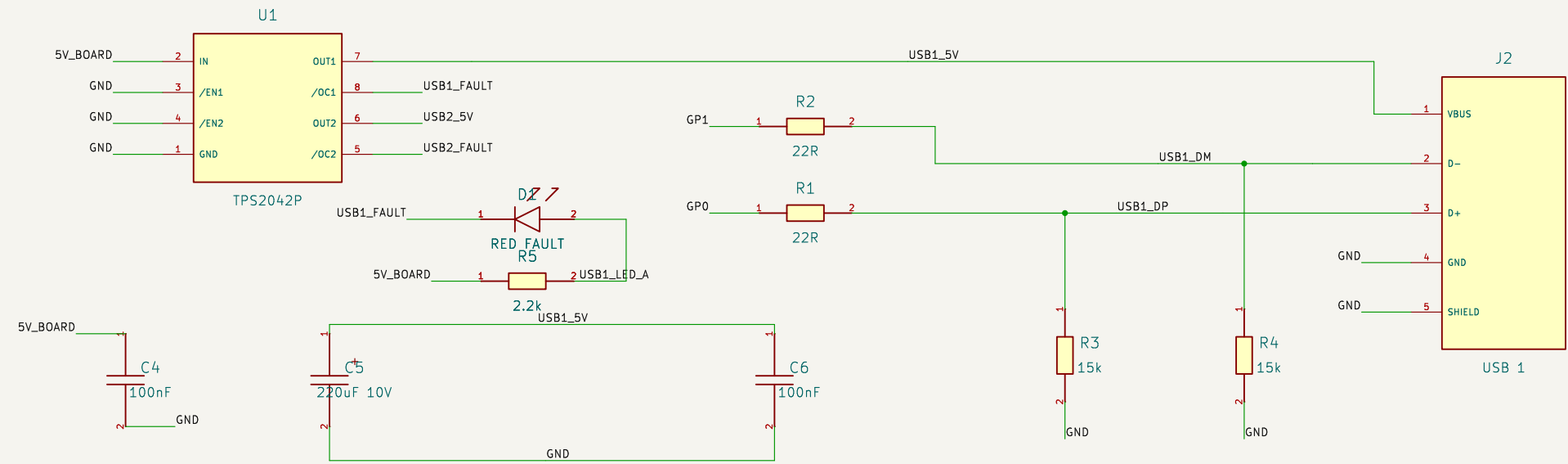
U2A-F are six buffers inside ONE CD4050BE chip.



Unused inputs go to GND.
Unused outputs stay open.

4 / USB KEYBOARD

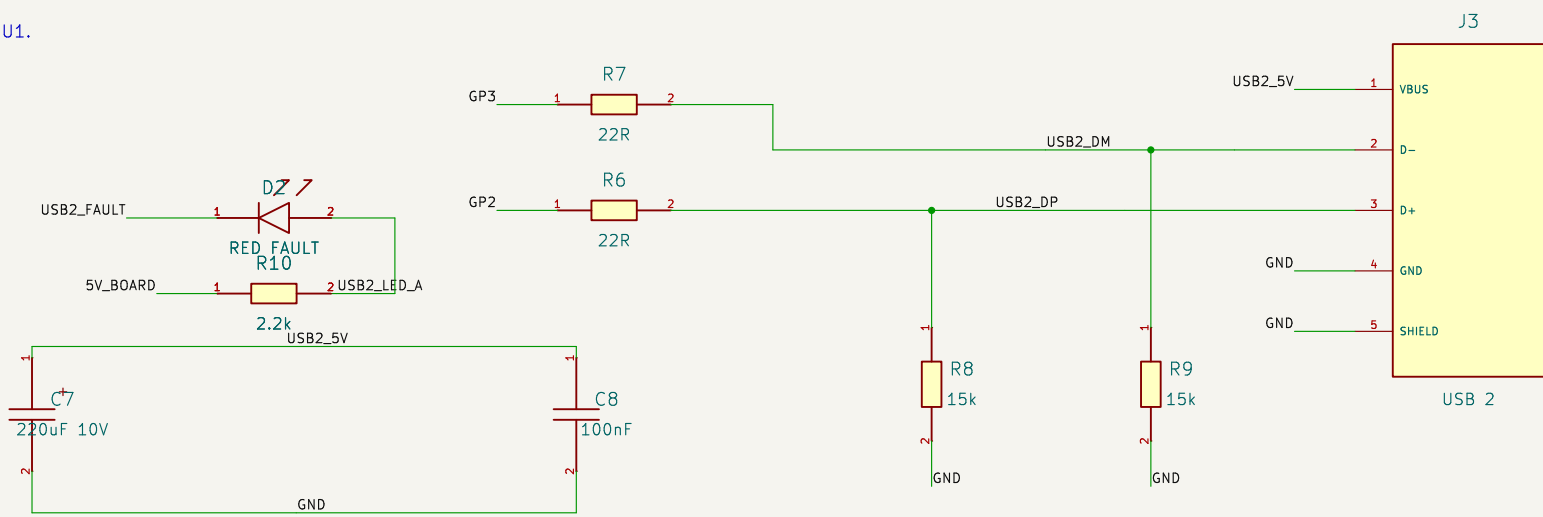
Use a supported USB device or its own 2.4 GHz USB receiver.



5 / USB MOUSE

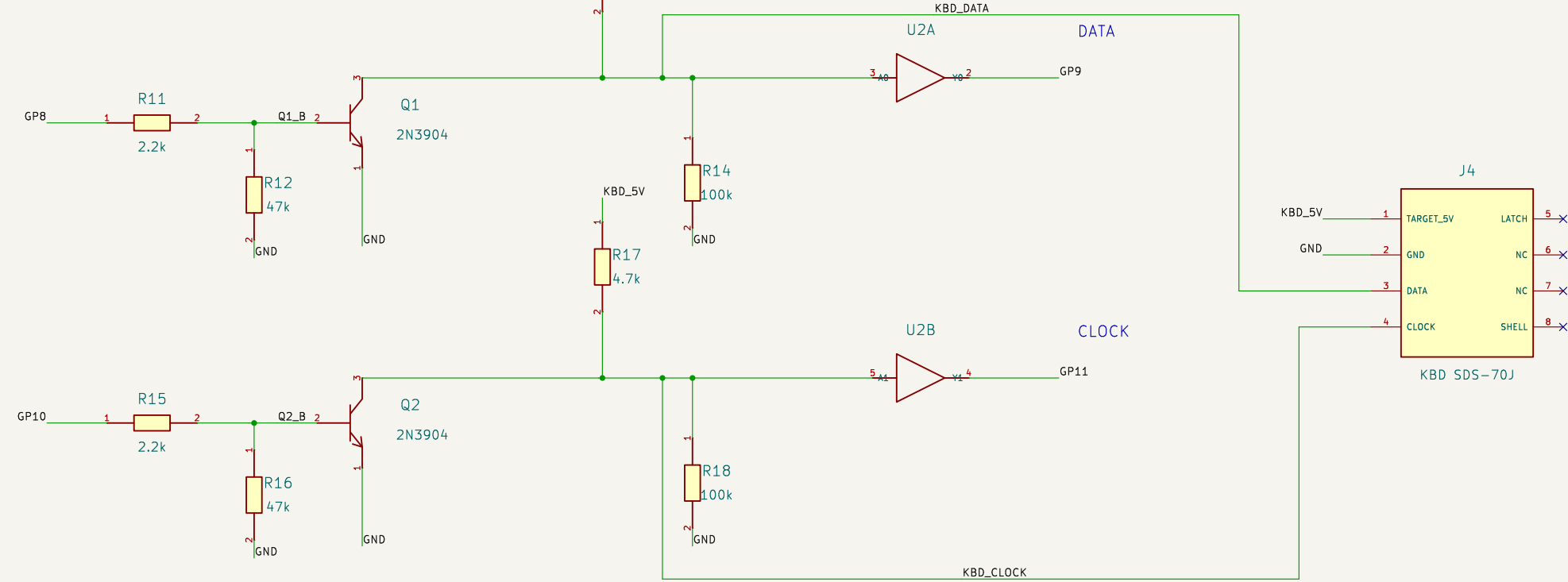
Use a supported USB device or its own 2.4 GHz USB receiver.

Power from U1 pin 6
Fault from U1 pin 5
Both channels are inside U1.



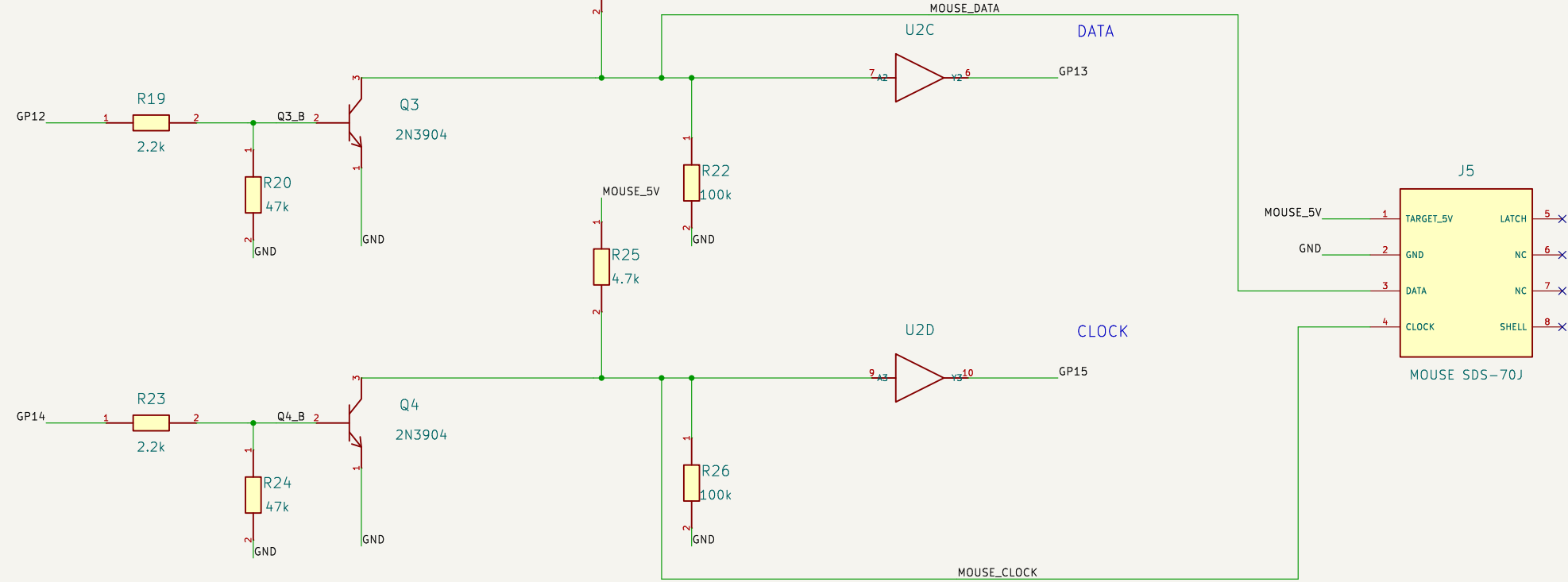
6 / KEYBOARD PS/2 OUTPUT

Read = cable level. Drive HIGH pulls the line LOW.



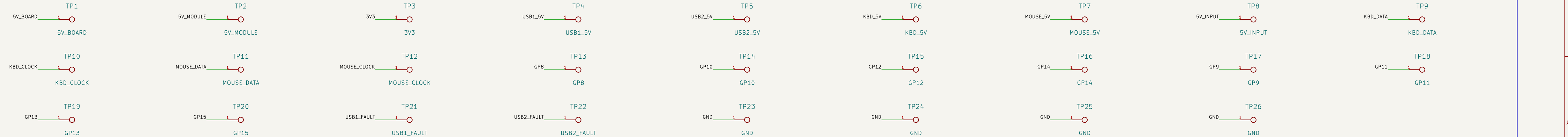
7 / MOUSE PS/2 OUTPUT

Read = cable level. Drive HIGH pulls the line LOW.



8 / TEST ACCESS

TP holes are optional probe contacts, not extra required parts. All GND test points are connected.



USB data: probe existing R3.1 / R4.1 (keyboard D+ / D-) or R8.1 / R9.1 (mouse D+ / D-). No added USB stubs.

HOW TO READ: dots join wires; crossing wires without dots do not join. Equal net names are connected.

Q = transistor, R = resistor, C = capacitor, U = chip, TP = test point, M1 numbers are carrier pins; use module GP labels.

Prototype: firmware and bench tests are required before use with the X16. See test-points.md and breadboard.md.